

Abstract

The Boolean satisfiability problem (SAT) is studied in this work from formal language theory. Within the line of research to which it belongs, the satisfiability of a formula reduces to the emptiness of the intersection of a regular language of interpretations with a grammar that enforces consistency among the occurrences of each Boolean variable. With less expressive grammars this reduction is limited to formulas with a certain order of occurrences: a context-free grammar admits no crossings, and with a simple range concatenation grammar (sRCG) the arity grows with the formula. In an attempt to solve more formulas in polynomial time, the problem is approached with range concatenation grammars (RCG): with their non-linearity the consistency of more formulas is modeled while the arity is kept minimal, as shown by an RCG that recognizes the consistency of any SAT formula with arity one; but with that same non-linearity the direct guarantee that the emptiness is polynomial is lost. A brute-force algorithm that decides this emptiness is given: the strings of the automaton are enumerated and recognized with the RCG; its cost is exponential, so enumeration is not a feasible route, but it shows why the problem is better studied on a single object. This object, the product grammar, is constructed, and its size and the cost of constructing it are characterized; on it the emptiness can be addressed: finding a polynomial method for all formulas, which would imply $P = NP$; proving that none exists; or finding one only by restricting the modeling to a subclass. Although its size is not polynomial, it remains an object of study: removing its non-productive clauses and analyzing the cost of deciding the emptiness are proposed as future work.

Keywords: boolean satisfiability, range concatenation grammars, emptiness of the intersection, finite regular language, non-linearity.